Part 1: Medical Electricity and Safety

1. Function of a defibrillator: It applies an electric shock to restore a regular heartbeat when the heart's electrical activity is in disarray (3:06-3:15).
2. Dry chest requirement: To ensure safety and prevent the electricity from traveling across the surface of the skin (4:11-4:14).
3. Ventricular fibrillation: A state where the electrical activity of the heart is in total disarray and blood is not being pumped (3:06-3:10).
4. Plastic handles: Plastic is an insulator, which prevents the user from receiving an electric shock (4:26).
5. Body as a circuit: The body conducts electricity, completing the path between the two paddles to deliver a shock to the heart (3:36-3:54).

Part 2: History of the Light Bulb

6. Platinum filament: It was too expensive (5:15).
7. Swan's material: Carbon (5:33).
8. Edison vs. Swan: Swan used a methodical, scientific approach; Edison relied on trial and error (5:24, 6:27).
9. Vacuum pump: To remove oxygen, which would otherwise cause the filament to burn up (5:08-5:13).
10. Tungsten: It replaced carbon because it can withstand high temperatures (up to 2,500°C) without melting (8:10-8:19).

Part 3: Electrical Circuits

11. Electric current: A flow of negative charge around a circuit (10:38).
12. Battery role: To give the charges energy and push them through the circuit (10:44).
13. Charges used up: No, electricity is not used up; the charges transfer energy to the bulb and continue moving back to the battery (11:05-11:18).
14. Instrument/Unit: Ammeter; Amps (11:39, 11:45).
15. Series circuit: Adding more bulbs increases resistance, causing the current to decrease and the bulbs to dim (12:43-13:01).
16. Series vs. Parallel: In series, components share the current and energy; in parallel, each branch receives the same current and potential difference (12:30, 17:23, 17:51).
17. Parallel brightness: They are just as bright as a single bulb (17:29).

Part 4: The National Grid and Demand

18. Electricity storage: It cannot be stored on a large scale, so supply must be matched to demand in real-time (14:00-14:14).
19. Spike factors: Commercial breaks (making tea) and weather/darkness (14:51-15:15).
20. Grid response: By predicting demand based on TV schedules and weather, they bring more power stations online as needed (15:22-15:34).